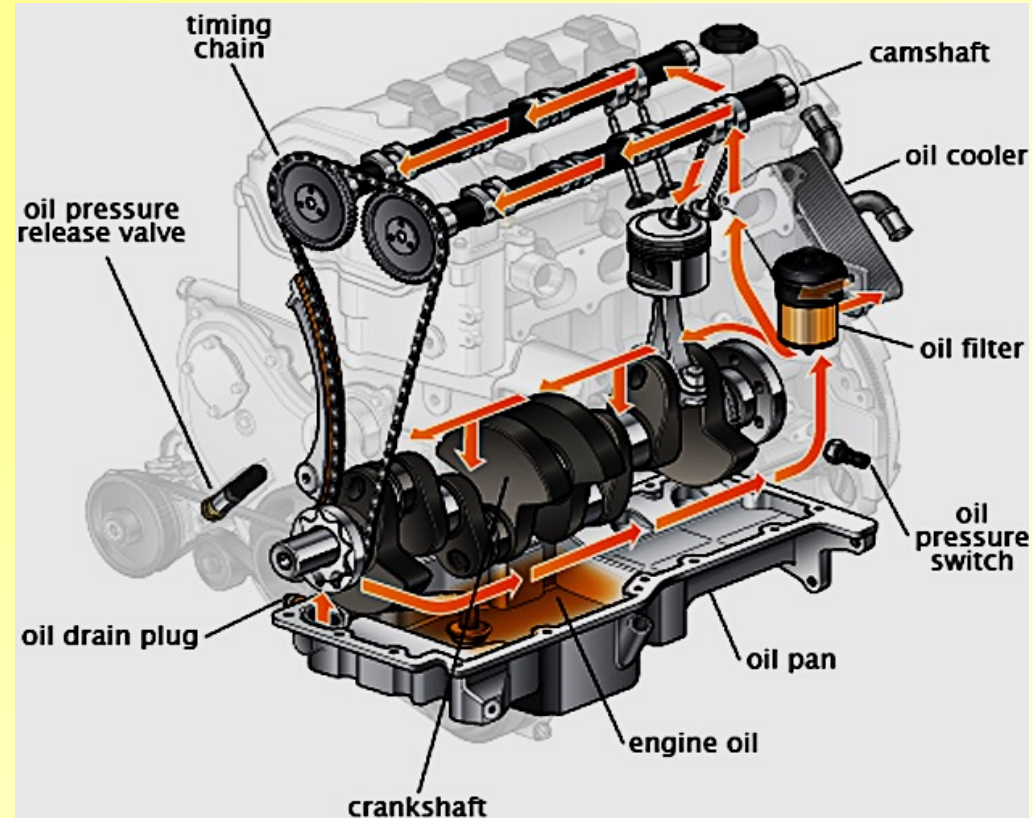




LUBRICATION OF I. C. ENGINES



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SYLLABUS



S.I. Engines: Deviation from the ideal cycle, combustion losses, flame development and propagation, effects of operating variable on detonation; Detonation rating of engine fuels, HUCR Dope, preignition combustion chambers, Mechanics of carburetion, MPFI systems, mixture requirement and combustion.

C.I. Engines: Stages of combustion, Diesel knock, effect of operating variables, knock rating of C.I. engine fuels, Cetane number; Effect of additives, combustion chambers. Requirements of fuel injection system and its mechanism, Fuel filters.

Lubrication and Cooling: Forced, splash, by-pass and direct, Blow by and crankcase dilution.
Air and water cooling, Thermostatic control.

Performance and Testing: High speed indicators, various efficiencies, Morse Test and Willan's line; torque and mean effective pressure, performance and heat balance sheet, Measurement of volumetric efficiency, Effect of atmospheric condition on performance of I.C. Engines; Supercharging; high altitude problem of I.C. Engines, methods and types of supercharging; suitability of S.I. and C.I. Engines;

Air Pollution: Vehicular emission, its norms, Causes and control of air pollution from diesel and petrol engines, catalytic converter.
Introduction to dual fuel and multi-fuel engines; free-piston engine and rotary combustion engine.



FRICTION



➤ *The work transferred to piston from gases contained in cylinder, is not available at drive shaft.*

- ✓ The un-available fraction of indicated work ($\sim 10\%$ at full load to 100% at idle) termed as friction work, is dissipated within the engine & its auxiliaries affect the max brake torque and min bsfc.
- ✓ A large part of friction losses appear as heat and must remove in radiator or oil cooler.
- ✓ Friction work, the difference between work delivered to piston while working fluid contained within the cylinder and usable work delivered to drive the shaft, can use to discuss friction,
- ✓ 1. **Pumping work W_p** : the net-work per cycle done by the piston on in cylinder gases during inlet and exhaust strokes.
- ✓ 2. **Rubbing friction work, W_{rf}** : the net work per cycle dissipated in overcoming the friction due to relative motion of adjacent components within the engine.
- ✓ 3. **Accessory work W_a** : the work per cycle required to drive the engine accessories, e.g. pumps, fan, generator etc.
- ✓ 4. **Total friction work W_{tf}** : is the sum of these three components.
$$W_{tf} = W_p + W_{rf} + W_a$$

➤ *The friction losses can be classified in two groups depending on type of dissipation which occurs.*

- ✓ 1. **Lubrication friction** between two metal surfaces in relative motion with a lubricant in between,
- ✓ 2. The friction work spent in **pumping fluids** through flow restrictions. The cylinder gases, cooling water and oil pump through the engine; fan pumps air over the engine block in turbulent mixing processes. Hence the frictional losses pumping will be proportional to N^2 .



LUBRICATION

➤ *In IC engines, parts move in / on each other causes*

- ✓ Friction between surfaces, wear-tear and rubbing heat generation;
- ✓ Friction reduces engine life, power output leads to performance reduction.

➤ *Lubrication is a process of applying a **substance** at correct pressure and temp between rubbing surfaces to minimize **friction** and allow **smooth** relative movement. The function of lubs are,*

- ✓ Oil film acts as a **Cushion** between metal parts, crank case oil temp ≥ 120 to 130°C
- ✓ **Reduce** frictional resistance, increase η_m and prevent wear-tear, protect against rust and corrosion
- ✓ **Cooling** (40% heat dissipation) piston & regions and **Cleaning** by control of dirt, contaminants
- ✓ Provide **Sealing** (hold gas & oil), reduces **vibrations/noise** level
- ✓ Power **transmission** (hydraulic machines) etc.

➤ *Lubricant requirement:*

- ✓ Stability against oxidation at 250°C of top ring, Detergency/dispersion control, wear reduction, Viscosity index: resistance against flow: high/ low fluidity, flash & fire points; Type of lubricants
- ✓ **Oils** : thin liquids, made of long polymer chains with additives (antioxidants, corrosion inhibitors & detergents etc.) viz, mineral oils (refinery), vegetable oils, animal oils etc.
- ✓ **Grease** : semi-solid, created by combining mineral oils with thickeners (lithium-based soaps, molybdenum disulphide, graphite etc.)
- ✓ **Dry lubricants**: solid, specific type - silicon, molybdenum, graphite, mica, PTFE etc.

- ✓ **Penetrating lubricants**: for un-sticking nut-bolts, removing chewing-adhesive stickers etc



FUNCTIONS & QUALITIES OF ENGINE OILs



S No	Main function	Where and when	Qualities required
1.	Reduce frictional resistance	During cold starting Between connecting rod/ camshaft bearings, pistons, rings and cylinder walls	Low viscosity to provide good pumping and avoid undue cranking resistance, Minimum viscosity without risk of metal to metal contact under varying conditions of temperature, speed and load; High viscosity at high temperatures; Good lubrication property beyond hydrodynamic conditions, especially at top centre; Antiseizure properties, especially during run in period.
2.	Protect against corrosion and wear	During shut-down or when running at low temperature In normal running	Must protect surfaces against corrosive action of fuel decomposition products (water SO ₂ , HBr, HCL etc. Must resist degradation (oxidation, thermal stability) Must counteract of fuel and lubricant decomposition at high temperatures, especially on non-ferrous metals; By intervention in friction mechanism must reduce the consequences of unavoidable metal to metal contact; Must resist deposit formations which would affect lubrication (detergency or dispersive action); Must contribute in elimination of dust and contaminants (dispersive action)



FUNCTIONS & QUALITIES OF ENGINE OILs

S No	Main function	Where and when	Qualities required
3.	Assist sealing	In the ring zone, especially at TC	Must have sufficient viscosity at high temperatures and low volatility, Must limit ring and liner wear, Must not contribute to formation of deposits in ring grooves and must prevent such formation
4.	Contribute to cooling	Chiefly of pistons, rings and con-rod bearings	Must have good thermal stability and oxidation resistance, Must have low volatility, Viscosity must not be too high
5.	Facilitate in elimination of undesirable products	During oil drains to eliminate atmospheric dust, soot from diesel engines, Pb salts, wear debris, organic products from burned fuel and lubricants and other contaminants which promote deposits or accelerate wear	Must be able to maintain in fine suspension all solid material (dispersivity) whatever the temperature, physical and chemical conditions (water), Must be able to solubilize certain organic compounds particularly heavy oxidation products



METHODS OF OIL LUBRICATION



➤ *Types of Lubrication Systems*

- ✓ **Mist** /charge lubrication
- ✓ **Wet-sump** lubrication
- ✓ **Dry-sump** lubrication

➤ *Mist or Petroil Lubrication*

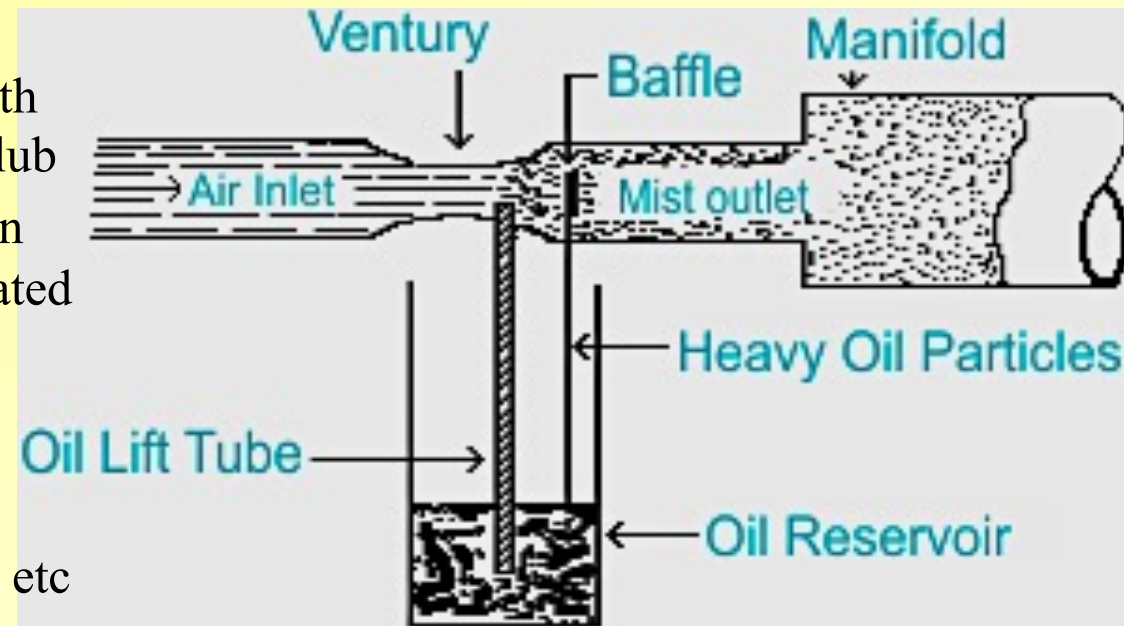
- ✓ It entails adding lubricating oil and petrol in fuel tank at a specific ratio (2 - 3%).
- ✓ Fuel + oil mixture induced in crankcase with air through carburettor for piston-cylinder lub
- ✓ In cylinder gasoline evaporates, oil stuck on cyl walls, keeps engine components lubricated

➤ *Advantages*

- ✓ Simple, economical, no maintenance cost
- ✓ No risk of oil leakage, system failure etc.
- ✓ Low engine weight due to abs of lub pump etc

➤ *Disadvantages*

- ✓ Carbon deposits & burning of lub oil-film
- ✓ Fouling of carburettor, spark plug, venturi etc.
- ✓ Limited use only in 2-stroke petrol engines





GRAVITY-FEED LUBRICATION SYSTEM

➤ Grease Lubrication

- ✓ Widely used being simple and sufficient for sealing
- ✓ There is no need of replenishment over a long time

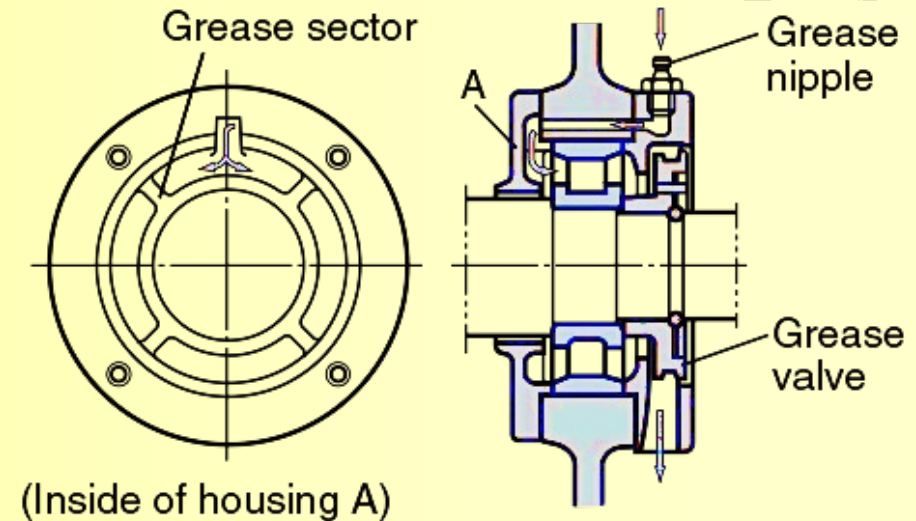
➤ Methods

- ✓ **Closed** lubrication - filled in advance in shielded/bearing housings
- ✓ **Feeding** method – sufficient quantity filled in bearing-housing and refilled at regular intervals
- ✓ **Central** method – by grease pump via. lines to points

➤ Oil Lubrication:

➤ Gravity feed system - Use gravity to feed oil into different machine's moving parts, engine points etc.

- ✓ A needle valve regulates and allow oil to flow by **gravity** from reservoir to respective wicked-parts.
- ✓ **Advantages:** gravity-feed systems are inexpensive, use less oil and easy in use
- ✓ **Disadvantages:** they result in oil loss, contamination
- ✓ They may not offer enough lubrication for high-speed engines for high-load applications.

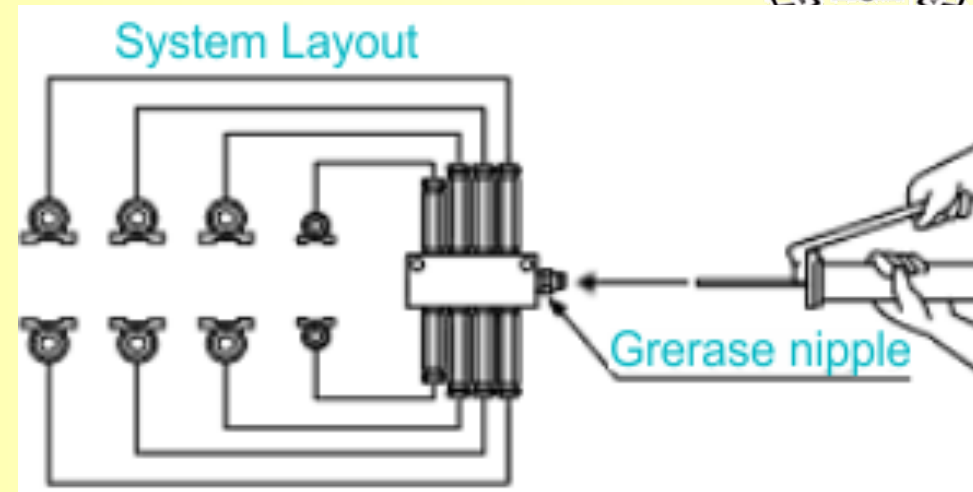




LUBRICATION SYSTEMS

➤ *Wet-Sump lubrication systems are;*

- ✓ *Splash* & circulating pump lubrication system
- ✓ *Splash & pump* feed lubrication system
- ✓ *Fully pump*/pressure feed lubrication system



S No	Item	Grease	Oil
1.	Sealing	Easy	Slightly complicated and required special care for maintenance
2.	Lubricating ability	Good	Excellent
3.	Rotation speed	Low / medium speed	Applicable at high / low speeds
4.	Replacement of lubricant	Slightly troublesome	Easy
5.	Life of lubricant	Relatively short	Long
6.	Cooling effect	No cooling effect	Good (circulation necessary)
7.	Filtration of dirt	Difficult	Easy



WET-SUMP LUBRICATION SYSTEM

➤ *In wet-sump lubrication, oil kept in tank at bottom of the crankcase known as oil sump, serves as oil cooler.*

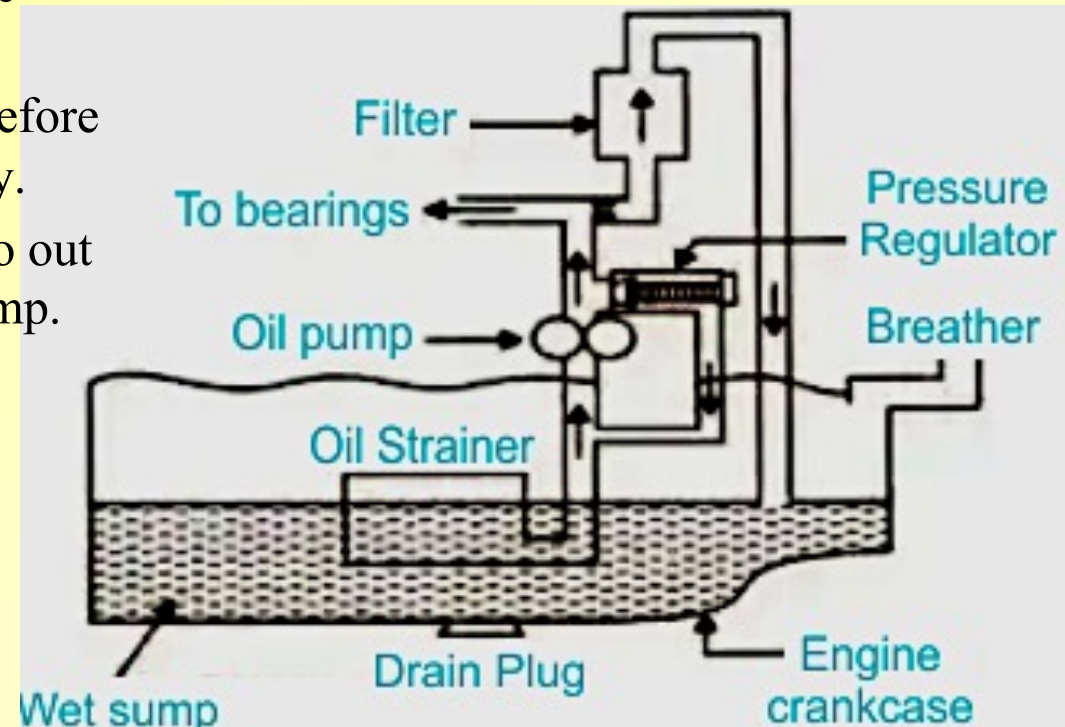
- ✓ Oil drawn from sump through strainer by a gear/ rotor pump
- ✓ Strainer prevents foreign particles/matter from entering into the oil circulation system
- ✓ Pressure relief valve is provided to prevent the excessive built up pressure of oil
- ✓ Pump supply oil to different parts of engine before draining back into the wet-oil sump by gravity.
- ✓ A breather allows oil fumes in crankcase to go out thus maintains atmospheric pressure in oil sump.

➤ **Advantages**

- ✓ Straight-forward, inexpensive and
- ✓ Doesn't require external hoses or reservoirs

➤ **Disadvantages**

- ✓ Elevated oil temperature
- ✓ Decreased ground clearance
- ✓ Oil hunger during high speed / power lubrication





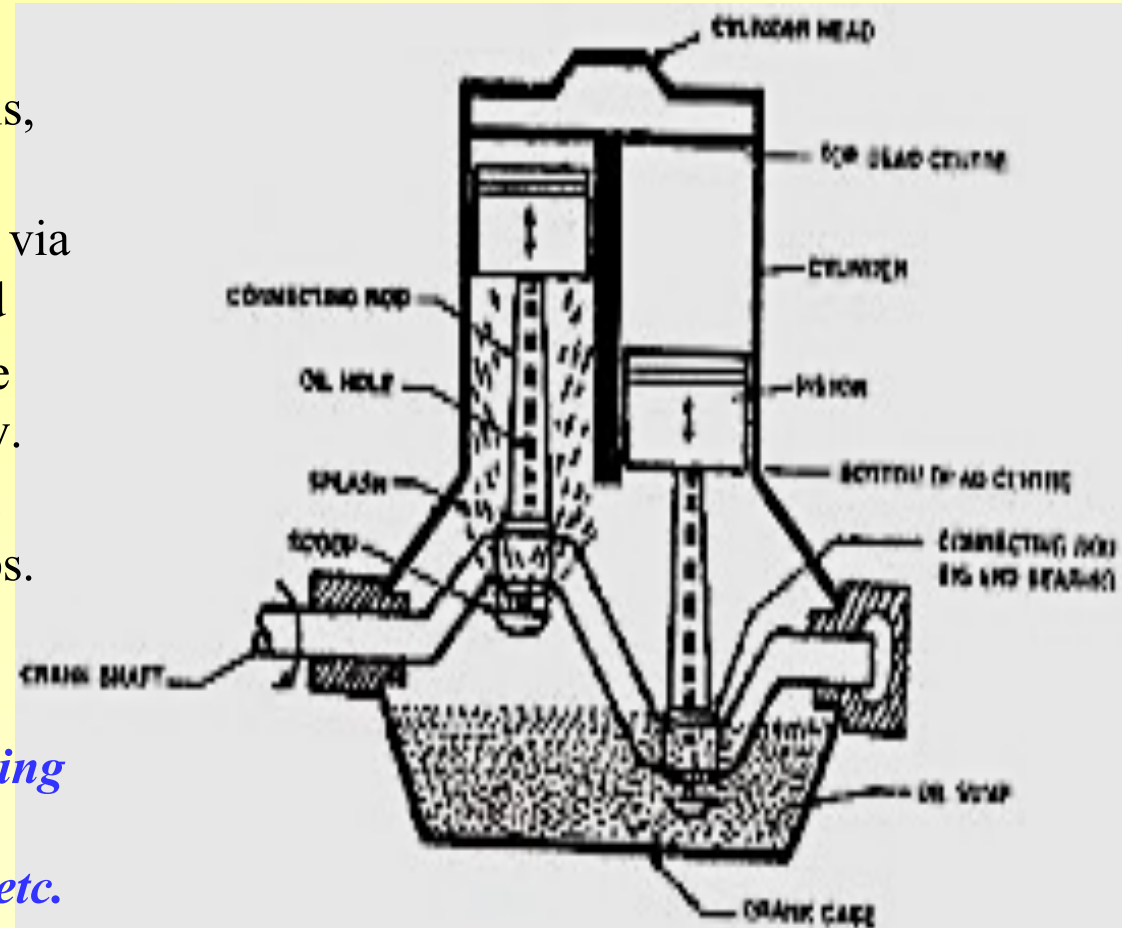
SPLASH LUBRICATION SYSTEM

➤ *Splash lubrication system*

- ✓ Lubricant filled in sump at bottom of engine
- ✓ Scoop/dipper fixed at end of connecting rod
- ✓ When piston moves to BDC, oil thrown by scoop from sump splashes on cylinder walls, piston, connecting rod, bearings etc.
- ✓ Crank-pin bearings receive oil from dipper via slots cut in lower end of the connecting rod
- ✓ Splashed oil stick on engine parts, lubricate them and returns back into sump by gravity.
- ✓ It also supplies oil under pressure via pipes oil-stream by dipper on connecting rod cups.
- ✓ This is appropriate for moderate-speed and loads, makes a thicker oil coat lubrication.

➤ *Splash system is not sufficient for higher bearing loads, hence oil at pressure is supplied by oil pump to parts, main and crankshaft bearings etc.*

➤ *Pump supplies oil under pressure to pipes which directs oil against dippers of connecting rod.*

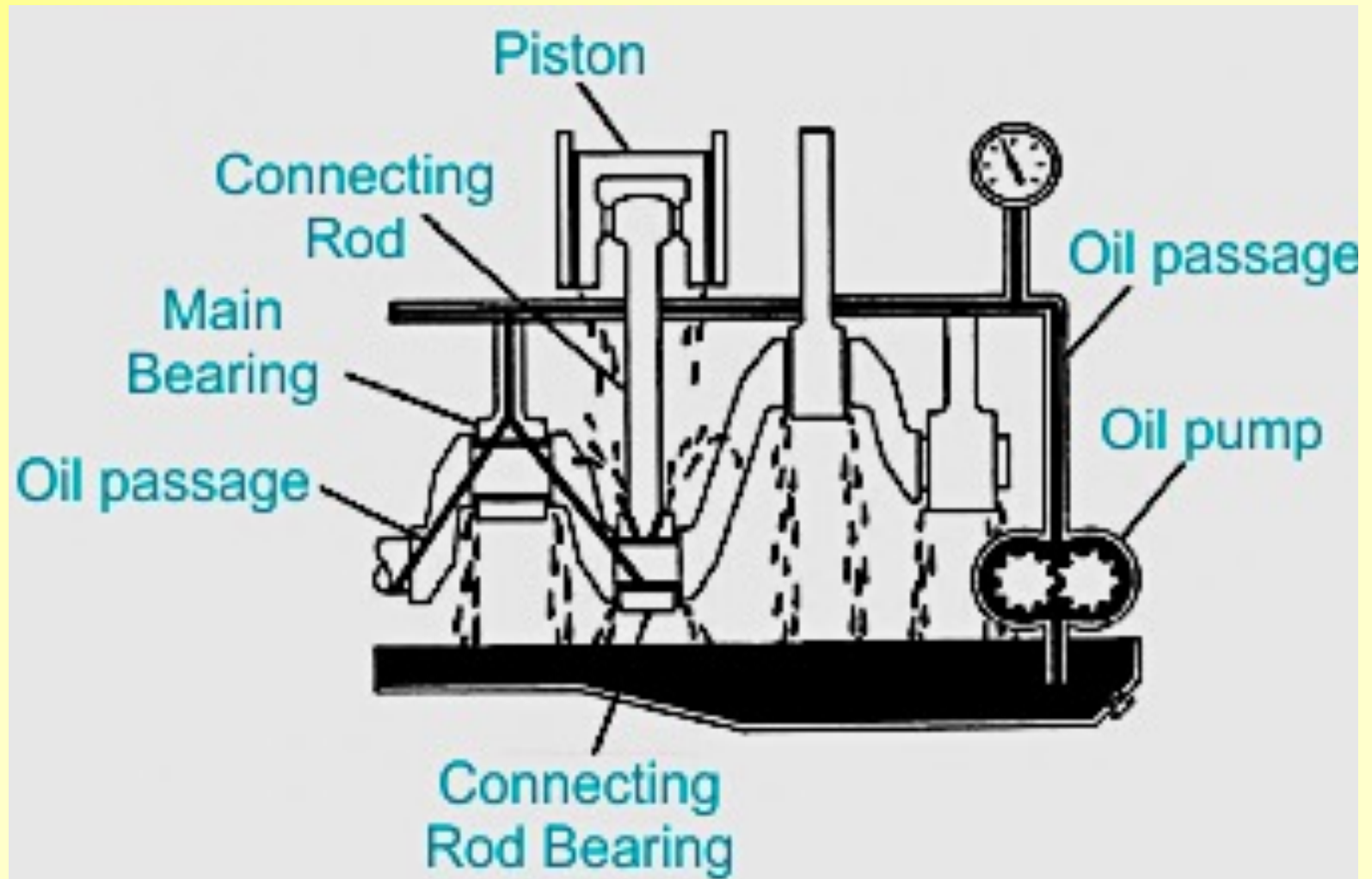


SPLASH LUBRICATION



FORCE-FEED LUBRICATION SYSTEM

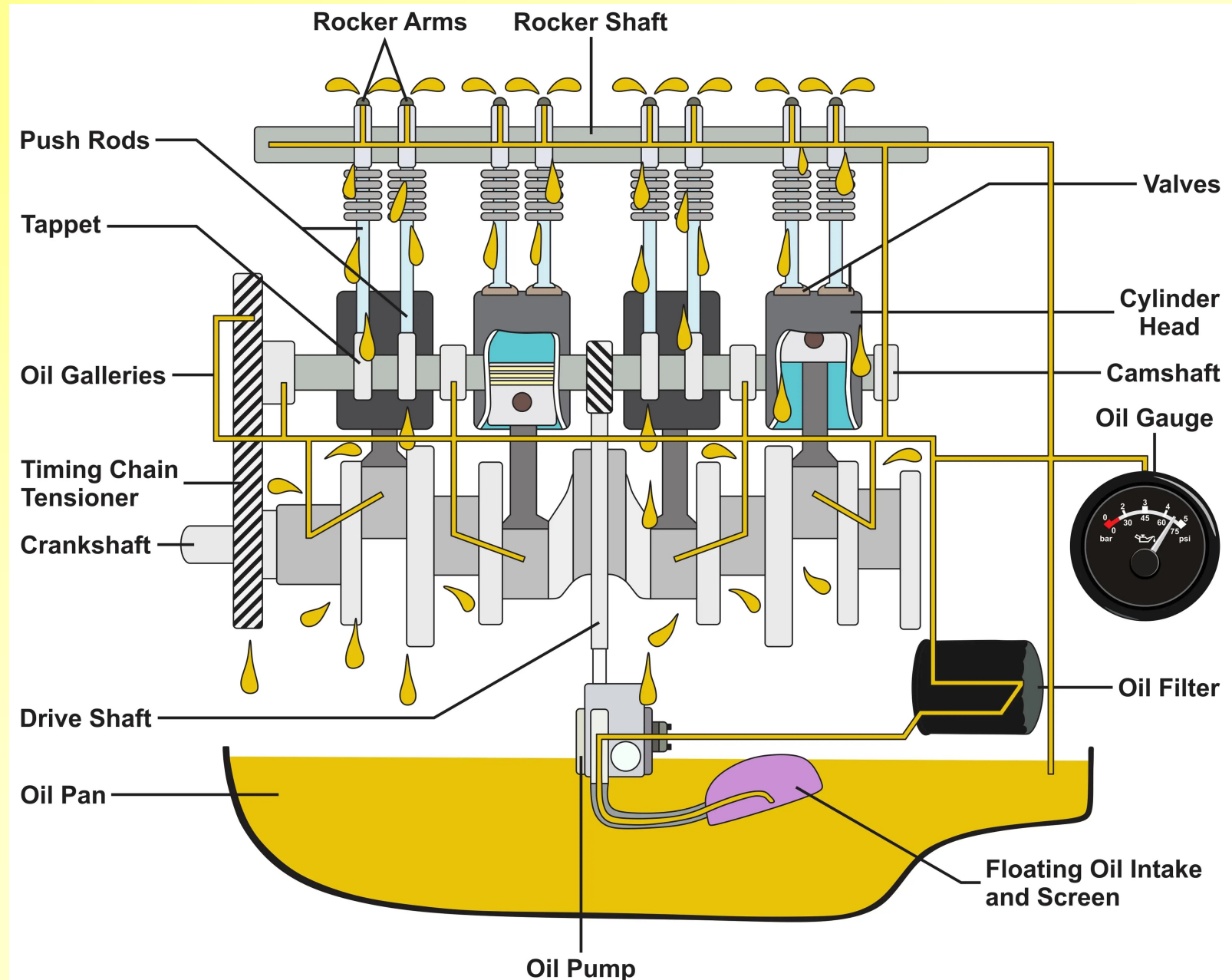
- *In this technique,*
 - ✓ Oil is forced through pipes or channels by an oil pump to the area that requires lubrication.
- *A relief valve or governor is used to control the oil pressure.*
- *This technique is appropriate for*
 - ✓ High-speed and
 - ✓ High-load applications for uninterrupted and consistent oil film maintenance.





PRESSURE FEED LUBRICATION SYSTEM

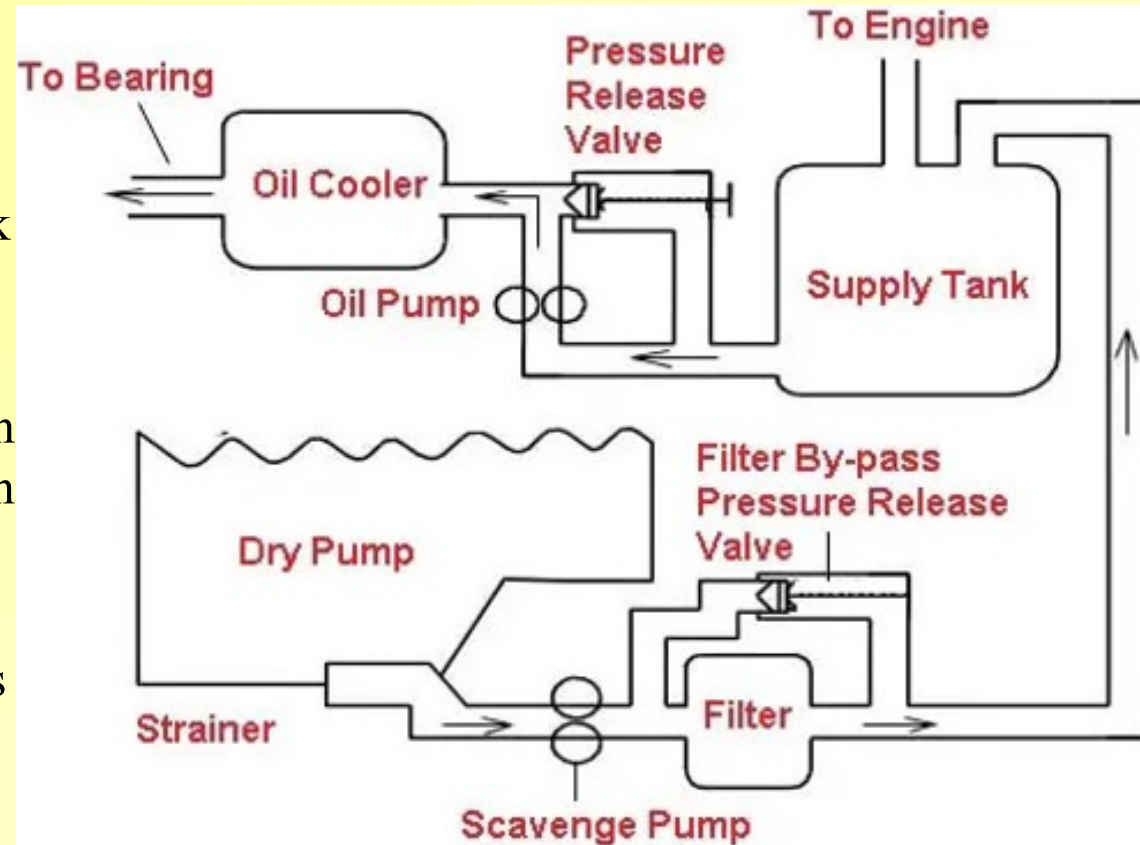
- In this, lubrication is done with the help of a pressure pump, submerged in oil sum
- After filtration, oil pumped to different parts of engine through oil tubes.
- From bearing oil floats to connecting rod through holes in connecting rod and camshaft.
- Then oil flows to piston pin, rings through holes and spread over the piston, rings, valves, cylinder walls and other parts where lubrication is needed.





DRY-SUMP LUBRICATION SYSTEM

- *Supply of oil is carried in an external tank.*
- *Oil kept in a separate tank behind radiator to cool and circulated in system by an oil pump*
 - ✓ Oil dripping from cylinders & bearings in sump is removed by scavenging pump, through filter & return back to supply tank
 - ✓ If oil filter **clogged**, pressure relief valve opens, permitting oil to **bypass** the filter.
 - ✓ Capacity of scavenging pump is more than oil pump; so prevent from accumulation in sump, keep dry so named as dry-sump.
 - ✓ Pump draws oil from supply tank & circulate under pressure to parts / bearings
 - ✓ A separate oil cooler, either water or air cooled is used to remove heat from the lubricating oil & greater system efficiency
 - ✓ Dry-sump are full force-feed engines
- *High-performance cars, large stationary engines use dry-sump lubrication system.*





BLOW-BY IN CRANK-CASE

➤ In reciprocating internal combustion engines, blow-by refers to the phenomenon of **leakage of combustion products** that escape/flow past the piston-rings-cylinder assembly into the **crank-case**.

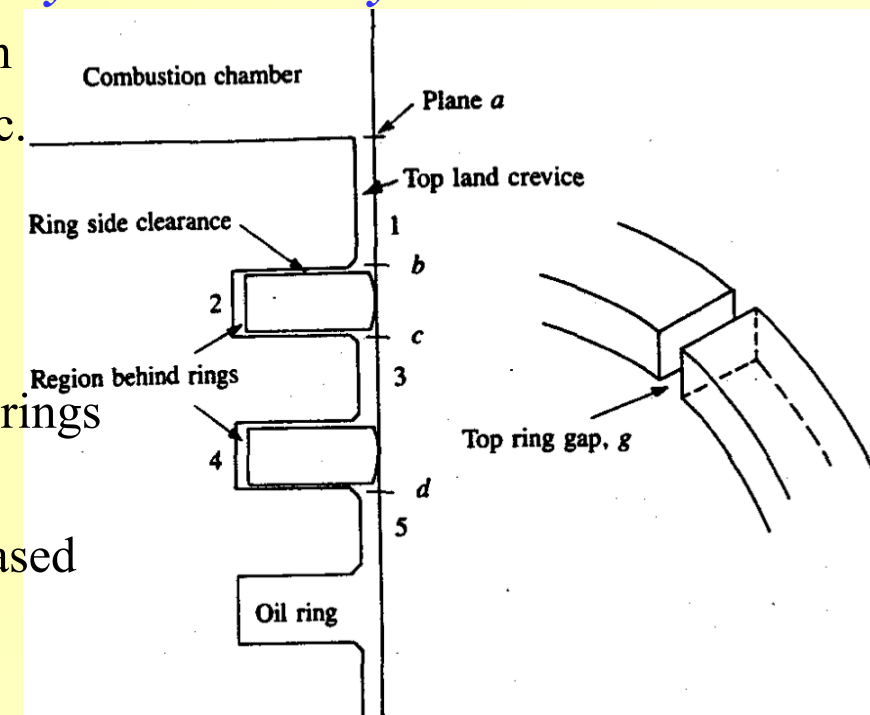
- ✓ Due to **high-pressure** in engine cylinders at combustion
- ✓ **Mixture** of air, burned/unburned HCs/gases, oil mist etc.
- ✓ Recirculated in engine to avoid external **pollution**.

➤ **Causes of blow-by**

- ✓ 1. **Wear of Piston rings**: fail to create proper seal,
- ✓ 2. **Wear of liner**: walls scored, inadequate sealing with rings
- ✓ 3. **Improper assembly**: lead gaps to allow blow-by
- ✓ 4. **High combustion pr**: turbocharged experience increased blow-by combustion pressures.

➤ **Effect of blow-by**

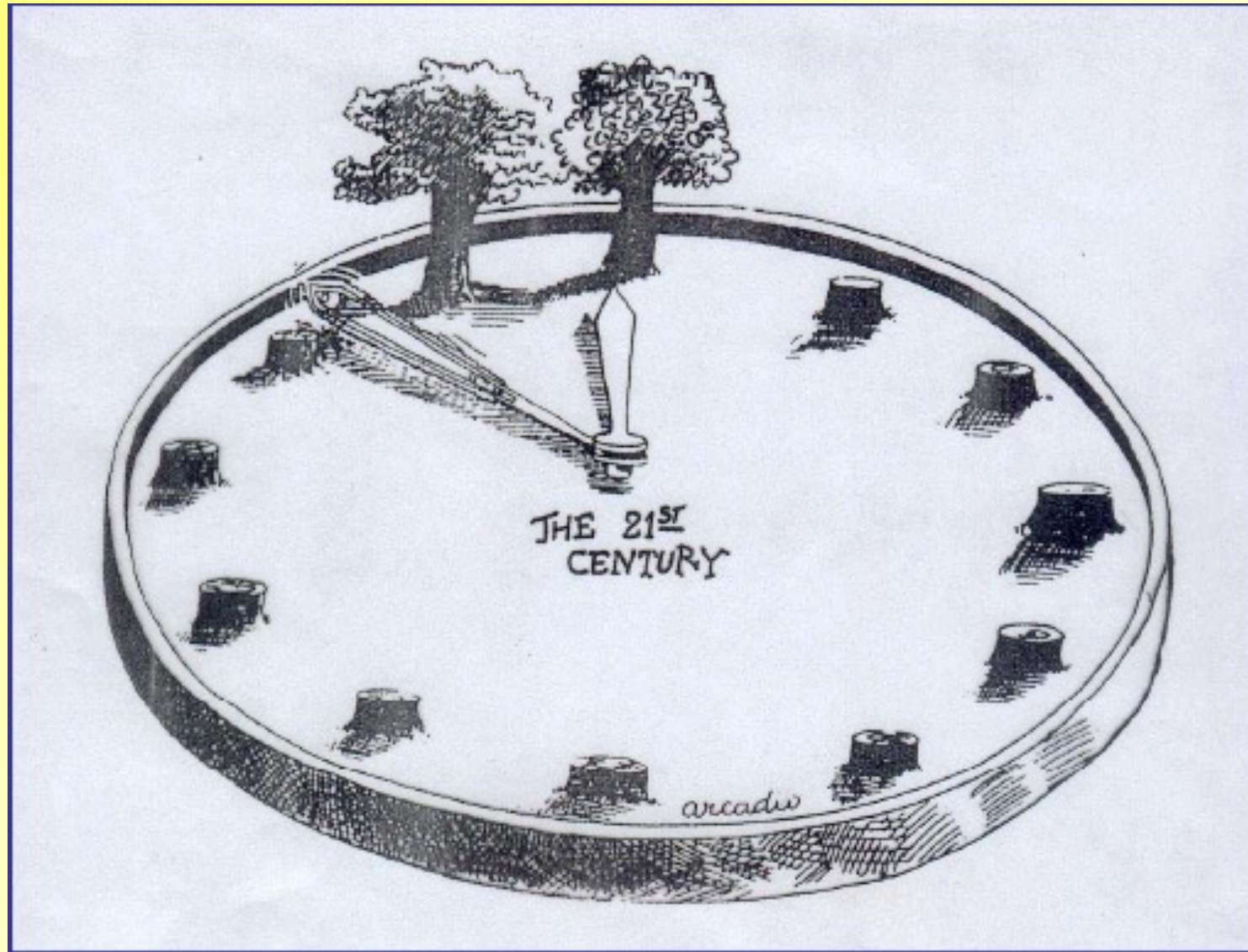
- ✓ 1. **Loss of power**: the escape of combustion gases, reduce engine efficiency and power output
- ✓ 2. **Increased crank-case pressure**: this can lead to oil leaks and may affect the operation of engine's ventilation system.
- ✓ 3. **Lubricant oil contamination**: combustion gases can mix with engine lubricating oil in crank-case leading to contamination and **reduced** lubrication effectiveness.
- ✓ 4. **Emissions**: increased blow-by can lead to higher emission of pollutants contributing to environmental concerns.





BLOW-BY & CRANK-CASE DILUTION

- **Crankcase Dilution:** it is a process where fuel / combustion products are leaked into the engine lubricating oil (5%) when they escape / flow past the piston rings and travel towards the crankcase.
 - ✓ This occurs because of most *seals / cylinders* are *not perfect* or
 - ✓ There is an opening through cylinder cross-hatches that causes *migration of fuel* and *exhaust gases* into the crank-case lubricating oil.
- **Causes of crankcase dilution**
 - ✓ 1. **Wet stacking:** when engine operate in *colder climates*, at lower operating temperatures, decreases combustion efficiency etc. which causes fuel to stuck on cylinder walls.
 - ✓ Once the fuel is stuck on walls / liner, it will *slowly* enter into the oil sump by being scrapping-off by piston rings and enter into the engine crankcase causes to dilute the lubricating oil.
 - ✓ 2. **Poor maintenance:** When oil contaminated with a small amount of fuel, it doesn't affect much the performance of oil or engine, but if added in large amount will affect performance; so regular maintenance, oil checks and changes can stop the dilution & any adverse effects on engines.
 - ✓ 3. **Crank-case ventilation:** cleaning of crank-case by *breather* or fresh-air fan to avoid dilution
 - ✓ 4. **Engine modifications:** Anything that affects the possibility of avoiding dilution of engine oil, manifests through *modifications* that produce a much greater amount of smoke passes into the crank-case before leaving via tailpipe, can avoid by introducing other types of oil contamination.
- **Effect of Crank-case Dilution**
 - ✓ Lowers the lubricating oil viscosity, enhances wear, lowers the engine performance etc.



Thank You